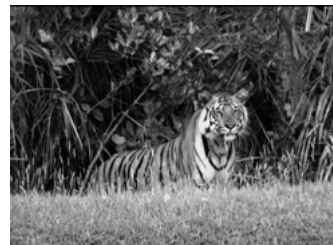


Chapter-5

The Rivers of Bangladesh and the Natural Resources

Bangladesh is a land of rivers. Rivers make the lives active in Bangladesh. Numerous rivers flow down the North of the Himalayas and the hilly regions of India through Bangladesh. These rivers move crisscrossing the country. Many of the rivers are so long and wide. Some are narrow. Many rivers are now non-existent and wiped off the map, and some of the rivers are on the verge of death. At present, Bangladesh has a total of 1008 rivers and rivulets. The total length of these rivers is 22,155 km. These rivers are our main resources. Besides, we have some land, soil, forest, agriculture and mineral resources including some other natural resources. The existence of Bangladesh largely depends on the accumulation, use, expansion and preservation of these resources. In this chapter, we will learn about our rivers and the natural resources of our country and will try to preserve this wealth.



After reading the chapter, we will be able to —

- describe the main rivers of Bangladesh such as the Padma, the Brahmaputra, the Jamuna, the Meghna, the Kornaphuli, the Tista, the Posur, the Sangu, the Naf and the Matamohuri, and to describe the sources, the routes of flowing and realise the importance of preserving them;
- explain the inter-dependence of rivers and dwellings;
- analyse the cause of water-crisis in a certain part of Bangladesh for the impact of the rivers of concerned countries;
- analyse the problems created by the water-crisis in some parts of Bangladesh and explain the opinion to resolve the issues;
- analyse the importance of rivers for communication, hydroelectricity and trade;
- realise the necessity of preserving the rivers to overcome water-crisis;
- explain the concept of natural resources;
- can compare the natural resources (mineral, forest, agriculture, water, and solar) of Bangladesh with several countries of South Asia (India, Myanmar and Nepal);
- explain the concept of water resource management;
- explain the importance of water resource management of water and food security of Bangladesh;
- describe the alignment of forestry of Bangladesh;
- analyse the significance of these resources in the economy of Bangladesh;
- be aware of preserving the natural resources.

Section 5.1 : The Rivers and the Water Resources of Bangladesh

The main rivers are the Padma, the Brahmaputra, the Jamuna, the Meghna, the Karnaphuli, the Tista, the Posur, the Sangu etc. Most of the known rivers of Bangladesh originated from the Himalayas, Tibbet, Borak of Assam and the hills of Lusai. All these rivers flow into the Bay of Bengal in the end.

The Padma: In India and in the North Bengal of India, the river 'Padma' is known to the people as the Ganges, and in Bangladesh, it is known as the Padma. It has its source in the Gangotri Glacier of the middle of the Himalayas. Flowing through several states of North India, the Ganges entered the district of Rajshahi of Bangladesh. The main flow of Brahmaputra has connected with the river Jamuna near Goalanda. It has united with the Meghna at Chandpur assuming the name Padma which has fallen into the Bay of Bengal flowing through Barisal and Noakhali. This is one of the largest rivers of Bangladesh, and the total area adjacent to the Ganges-Padma is 34,188 square kilometer. Of the numerous branches of rivers flowing from the East to the West, the mentionable are the Bhagirathi, the Hugli, the Mathabhanga, the Isamoti, the Bhoirab, the Kumar, the Kapataksah, the Naboganga, the Chitra, the Modhumati, the Arial Kha, etc.

The Brahmaputra and the Jamuna: The Brahmaputra originated at the Manos Sarobar (a large lake) of Tibbet. It has entered the district of Kurigram of Bangladesh flowing through Assam. Before 1787, the main stream of river Brahmaputra flows through Mymensingh from the northwest to the southeast corner. But in 1787, the bed of the river rose up due to the earthquake, and the bed lost the containing capacity of water. So a new stream was created and it was named as the Jamuna. It was known as the Jamuna up to Goalanda to the South. Being united with the Ganges, it assumed the name Padma. The river of Dholeswari is a distributary of the river of the Jamuna. The river Buriganga is a distributary of Dholeswari. The Jamuna's tributaries are the Dhorola, the Tista, the Karatoa, and the Atrai rivers. Up to the confluence of river Ganges, the length of the Brahmaputra is 2897 km and the length of the shore is 5,80, 160 square km of which 44,030 square km is situated in Bangladesh.

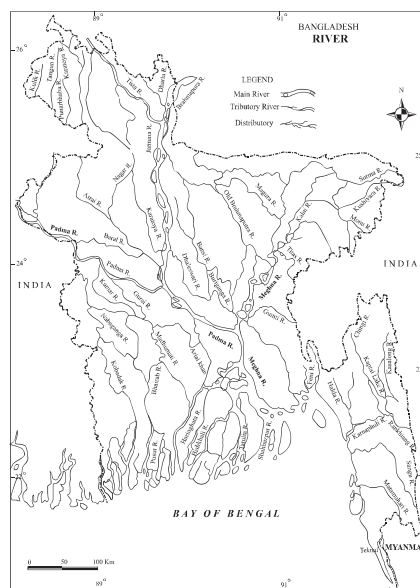


Figure 5.1: The rivers of Bangladesh

The Meghna: The Meghna has been created at the confluence of the river Surma and Kushiara in the district of Sylhet. Both the rivers originated from the river

Borak of Asham in Naga-Monipur zone, and entered Sylhet of Bangladesh. It was named the Meghna, and moved towards the southwest corner after assuming the name Kalni near Azmiriganj of Sunamganj. Passing through Bhairab Bazar, it is united with the old Brahmaputra. The confluence of the Buriganga, the Dholesori and the Shitalaksha has connected with the Meghna, and meeting with the Padma near Chandpur, the river has created an estuary. In the end, it falls into the Bay of Bengal. The river Monu, Titas, Gomoti, Bowlai are the branches of the Meghna. The alluvium soil and flood during the monsoon make the land of Bangladesh more fertile.

The Karnaphuli: The main river of the southeast region of Bangladesh is the Karnaphuli. Its inception point is the mountain of Lusai. The river stretching the length of 320 Km falls into the Bay of Bengal flowing near the Chattogram city. The main branches of the Karnaphuli are river Kaptai, Halda, Kasalot, and Rangkhia. Chattogram, the main sea port of Bangladesh, is situated on the bank of the river Karnaphuli. This river is important for the hydroelectric power station and the Chattogram port.

The Tista: It originated in the hilly area of Sikkim, and flowing through Jolpaiguri and Dargiling of India, the Tista entered Bangladesh through the Dimla upazila of Nilphamari. It changed its direction because of the flood in 1787 and started flowing through a dried up stream of the Brahmaputra. Before changing the direction, it was connected with the river Ganges, but now it is flowing together with the Brahmaputra. At present, the length of the river is 177 km and the wide ranges from 300-550 meters.

The role of the Tista in draining away water in the northern region of Bangladesh is vital. The Tista barrage project had been constructed in 1997-98. The barrage is supposed to play a significant role in water conservation, drainage, irrigation, and flood prevention in the region, however, Bangladesh has not yet gotten its full benefits for various reasons.

The Posur: The Bhairab or Rupsha, flowing by the south of Khulna, advanced further and by the right side of the landmasses of Trikona and Dubla and through the southern Sundarban of the Mongla port has fallen into the Bay of Bengal. The length of the river is 142 km, and the width ranges from 460 meters to 2.4 km. This is so deep that the ships can either enter or leave the Mongla sea port easily throughout the year. The Posur plays an important role as a river-route of Khulna to Barisal and vice versa.

Individual work: Identify the impacts of the rivers on rural economy.

Individual work: Identify the causes of low tide of the rivers.

The Sangu, Feni, Naf, Matamuhuri : Having originated from the Arakan hills and entered through Thanchi of Bandarban district, it was named 'Sangu'. It has fallen into the Bay of Bengal. It is 294 km of length. On the other hand, the river

Feni had its origin in the hilly area of Tripura, and entered the district of Feni. The river Feni has fallen into the Bay of Bengal flowing by the north of Sandeep. The river Naf is situated at the border area of Bangladesh and Myanmar. Its estuary is very wide. It has fallen into the Bay of Bengal. Its length is 56 km. On the other hand, the river Matamuhuri has its source in the Maiver hills of Lama. This river has fallen into the Bay of Bengal by the western side of the district of Chokoria. Its length is about 120 km.

The influence of these rivers on the economy of Bangladesh knows no bounds. Those are the sources of irrigation, industrial use and the generation of hydroelectricity. Not only that, the rivers are the sources of fish and proteins too. Those are the main routes of transportation and communication too. These rivers carry the silt to increase the fertility of the land. The importance of these rivers is immense in our economic development since the agriculture, industry, trade and business, transportation, etc. largely depend on the river. The flow of our rivers has been feeble, water is being polluted and the navigability is lost for creating obstruction against the flow, throwing industrial waste, maritime waste, untreated sewerage, etc. We all should be aware to protect these rivers from being destroyed.

The Reciprocal Relation between the River and Human Habitation

Human beings started living on the riverbanks since the ancient time because rivers would ensure their necessary water. On the other hand, water for irrigating cultivable land was also possible to get from the rivers. Rivers are closely related with catching fish for survival. Their roles as the main sources of food and earnings are also significant. Rivers have a great contribution to the building of human civilization. Subsequently, human beings used rivers for the development of life and livelihood. They would establish their dwellings near the rivers and search for livelihood too. As a result, the relation between rivers and the human being has been reciprocal. At present, this relation has been multifarious and close.

Human beings have set up the villages and towns as the permanent habitation through catching fish, carrying goods, and developing trade and commerce, etc. depending on the rivers. The rivers have become water resources. The population growth was utmost on the riverbanks. So, most of the towns and cities are built-up on the banks of the rivers such as Dhaka is built on the bank of the river Buriganga, Chattogram on the bank of the river Karnaphuli, Narayanganj on the bank of the river Shitalakksah, Sylhet on the bank of the river Surma, and Cumilla on the bank of the river Gomoti.

Rivers are most useful to the industries and factories, etc. is utmost. Rivers are given priority for developing a modern irrigation system to cultivate land by the modern technology. It is planned to bring the vast area of Kustia, Jessore and Khulna under the irrigation system from the project of the Ganges-Kapotaksha project. The people of that area will be benefited in agricultural production. From the multifarious project planning of the Karnaphuli, ships are being moved through the waterway of 644 km, cultivation is being possible in 10 lakh acres of land, and hydro-electricity is being generated building a dam at Kaptai. As a result of this dam, it is largely possible to keep the Chittagong region free from severe floods, but this dam has many negative effects on the lives of the people living in Rangamati, such as eviction from the land, permanent conflict between different ethnic groups, etc.

The people of Rangpur, Bogura and Dinajpur are getting benefit from the Tista barrage project. The improvement of farming in the greater Cumilla, Noakhali and Chattogram is becoming possible by drawing water from the river Meghna. Various agricultural projects have been taken using the river water. As a result, the agro-economy of the country is being developed day by day, employment of human being and food security are being ensured. In respect of communication with the people of the southern zone of the country through the waterway, the number of water vessels are being increased day by day. River route is selected for the transportation of goods. It is also chosen to give India the transit. As a result, Bangladesh gets the opportunity of earning a huge amount of foreign currency. The role of rivers is being increased day by day to develop the economic condition of the people of Bangladesh, to protect their health, to ensure fresh air and the supply of water in the cities. In the northern zone of the country, as the rivers are being dried up, crops fail, and the human life is facing the threat of destruction. That is why, dredging is needed to keep the navigability of the rivers immediately. All the rivers of the country should be protected to develop the economy, and the rural as well as the urban life. For this reason, the ecologists are carrying on awareness programme like 'save the rivers.'

The Reasons for Water Crisis, its Impact and Steps for Solution

The Reasons for Water Crisis

Despite numerous rivers, streams, canals, rivers, and streams in Bangladesh, water scarcity is becoming acute in several regions. Although the causes of water scarcity vary from region to region, there are some common causes. A lot of water comes from upstream in the rivers of Bangladesh. There is a lot of silt in this water. These sediments are deposited at the bottom of the river. For a long time, several rivers have been filled with silt. As a result, water flow has decreased and siltation increased in the rivers. Moreover, many rivers have disappeared. In many districts in the northern and southern regions, many such

rivers become familiar as dead rivers. It is necessary to excavate the sediments deposited at the bottom to maintain the regular flow of the rivers. However, the dredging efforts are not so strong. As a result, water flow and navigability of many rivers have been reduced. This is to remember that frequent excavations in the river may be harmful to animals and plants living in the river.

In the dry season, the Tista, the Padma, and the Kapotakkho have been dried up. The north-west zone of the country faces acute water crisis for constructing dam at Farakka. All the rivers including the Padma face negative consequences. The water crisis causes an ecological imbalance especially in the dry season.

The flow of water in the main stream of the rivers is being low for pumping out a huge quantity of water for irrigation and other essential works. So, the rivers have lost their

Group Work: Make a list of the problems created by the low tide of the rivers.

usual course and complexion. Many rivers are being dried up for constructing bridges, culverts, and dams disregarding the rules and regulations.

The rivers are gradually losing their navigability in the dry and winter season for low tide, and it has an adverse effect on the society, economy and human life. Trade, fishing, communication are almost being closed. Those who dwell on the banks of the rivers are compelled to leave in search of different livelihood.

The tide of the rivers becomes low in the dry season, but high in the rainy season. As a result, the dwelling places of the people are washed away because of tremendous pressure of the water, and people become homeless and destitute. If the rivers are dried up, there will be crisis of fish, and malnutrition may occur for protein deficiency.

If the rivers lose their flow, the trade and profession that have been created a long ago may be destroyed, and the people who earn a living from the trade may be poor and destitute. The greenery on the bank of the rivers may be destroyed without water. So the existence of human beings and animals may be endangered. So, the rivers should be dredged regularly to keep the usual flow and unnecessary barrages, bridges, and culverts should not be built over them.

The Role of the Waterway in Communication, Hydro-electricity and Trade

The importance of the waterway in the fields of communication, hydro-electricity generation and trade is mentioned below:

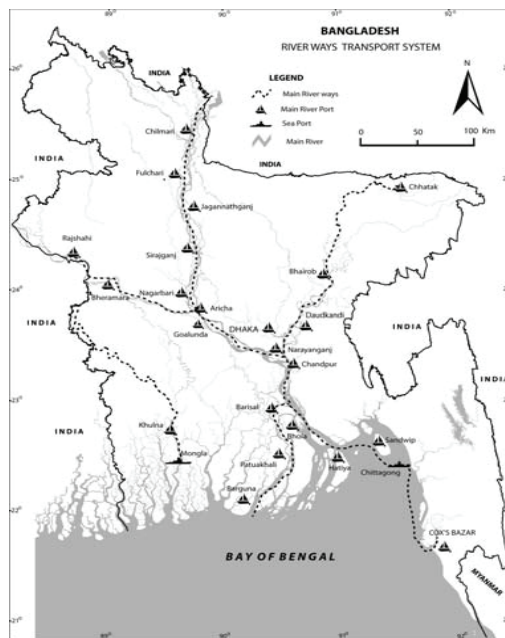
The Communication: The Rivers play a significant role in communication systems in the land of rivers. The Padma, the Meghna, the Jamuna, the Karnaphuli, the Kushiara, the Matamuhuri, the Atrai, the Modhumati, the Gorai, the Kirtankhola, the Karatoa, the Titas etc. rivers play an important role in providing service of carrying the passengers. All consider the waterway as the busiest route. The length of the waterway of this country is about 9833 km of which 3,865 km remains busy with the movements of marine vessels. Through the waterways of this country, millions of people travel by boats, launches, steamers, ships, etc every year.

The Hydro-electricity: The electricity that is generated by using either the current of river-water or waterfall with the help of a turbine is called the hydro-electricity. It is a renewable power resource. The hydro-electricity started generating power during the Pakistani period by constructing a dam across the river Karnaphuli at Kaptai. This electricity can be produced at a very low cost. In the present world, the cost of electricity which is produced by oil, gas or nuclear burner is very high.

As the generation cost of the hydro-electricity is very low, the increase of the production of electricity by using water resource is more profitable for the economy of Bangladesh. But, the mountains and hills convenient for generating hydro-electricity are few in number in our country. That is why Bangladesh has a few scope for producing hydro-electricity.

Trade : A total of 75% commercial goods are transported through the inland water way in Bangladesh. Now the number of cargo ships are being increased day by day. Goods are being transported through almost all the waterways by the government and private enterprises. Goods can be transported easily even though unrest prevails in the country.

In the rainy season, most of the goods transports are run by the waterway. But in the dry season, due to the navigability problem, the movement of ships in some of the rivers is rapidly decreasing. There is no alternative of water transport to develop our agriculture, fishery and industry. All the government and non-government institutions and organizations should come forward to make long-term planning and implement afterwards to make the trade and commerce of Bangladesh forceful. Then Bangladesh will be capable of achieving economic solvency by using the facilities of waterways and water resources.



Picture 5.2 : Map showing communication system of Bangladesh.

Group work: Make a list of the activities that depend on the rivers.

Section 5.2 : The Natural Resources of Bangladesh

The Concept of Natural Resources

The goods that are obtained by the nature are usually called the natural resources. Those are soil, water, forests, solar energy, fish, minerals, etc.

The Natural Resources of Bangladesh and Several Countries of South Asia

The natural resources of Bangladesh are almost same in comparison with the resources of several South Asian countries like India, Myanmar and Nepal. The descriptions of these natural resources are given below:

The Agricultural Resources: The South-Asian countries like India, Bangladesh and Nepal are agro-based in nature. The farmers of these countries grow agricultural products depending on the land, nature, climate, soil, and rivers. Rain and temperature of a certain degree are needed to grow these agro-products. As a result, production varies in different areas. In some areas of India, the production of crops hampers either for immense cold weather or for scanty rainfall. At the bottom of the Himalayas of Nepal, crops grow very little, but in the river washed fertile zone of West Bengal of India, Bangladesh and Myanmar, paddy, wheat and other crops grow several times in a year. Paddy, potato and wheat grow plenty in Bangladesh. In the eastern zone of India and in the hilly area of the northeast zone of Bangladesh, tea is being produced. The production of wheat, maize, mustard, etc. are plenty too. The soil plays an important role in producing these crops. Cotton, tea, pepper etc. are grown well.

The Forest-born Resources: The relation between the forest and the climate is very deep. The climate of Bangladesh is different from that of the South Asian countries like India, Myanmar and Nepal. In the hot and moist climatic zone, deep and extensive forests grow because of heavy rainfall. In the hill tracts of Chittagong, the eastern zone of India and the northern zone of Myanmar, the ever green forests have been developed.

The Fisheries: The land and the climate have a direct relation with the fisheries of a country. Bangladesh is regarded as a country of fishes for having rivers, canals, ponds, pools, marshes and fens in plenty. Many kinds of small and big fishes are found here. The Bay of Bengal is a big source of fishes. The neighboring countries of Myanmar, India and Nepal have sufficient fish resources.

The Mineral Resources: Various kinds of mineral resources such as gas, oil, limestone have been found under the ground of hilly districts of the eastern zone of Bangladesh and coal and limestone have been found in northern region. The demand of gas of the country can be fulfilled by collecting these resources. Natural gas has been found at the bottom of the sea. Still there are the possibility of having many kinds of animal and mineral resources. India is a large country. It is enriched with various mineral resources for having geological varieties. Myanmar has mineral resources in abundance regarding mineral resources, but Nepal is still lagging behind.

The Solar Energy: The sun emits rays Perpendicularly almost throughout the whole year in the southern latitude of the equatorial zone. So, Bangladesh and the other countries of the South Asia gets a huge quantity of solar energy easily for being situated in the equatorial or tropical zone. In the countries of this zone, the temperature never falls down. So the people never have to stay in dark. In several countries of Europe and American continent, the sun gives rays in a slanted way. Sometimes the sun cannot be seen even.

Group work: Make a list of the natural resources mentioning the important roles on different fields of the economy.

Group work: Prepare a chart mentioning the steps that should be taken to preserve the natural resources of Bangladesh.

That is why the people of those countries have to spend a huge quantity of fuel to keep their residence suitable for living. The countries of our zone have not to do so. The sunlight that we get easily from the nature is a very valuable solar resource. We can use this solar energy through modern technology, and fulfill the demand of electricity. The possibility exists that we can develop in the fields of food, health care, and residential facilities by using the solar energy.

The Concept of Water Resource Management

The role of water is important for the existence of the animal and the human beings. So, water is a very valuable resource. The role of water is immense in the development of agriculture and industry. Though we can get sufficient water from the rainfall, the agricultural and industrial productions are hindered and life becomes complicated for the want of water in the cold and dry season. For that reason, the flow and distribution of water should be ensured through the proper management. Getting and using water in a planned way is called the water management.

Individual work: What are the efforts that might be taken for water and food safety of your locality?

Water usually exists in the solid, liquid and steam form. In the cold and dry season, the use of water can be ensured through proper management by reserving the water in the rivers, canals, ponds, pools and marshes. In the modern age, the management of water resource has been emphasized for the welfare of human beings, or the misuse, scarcity, and toxicity may cause environmental disaster and existence of the flora and fauna may face the threat of extinction.

The Importance of Water Resource Management for Water and Food Security in Bangladesh and Peoples' Involvement

Bangladesh is the most densely populated country of the world. It is hard to ensure the food, water and other securities of this large population. So, it has been difficult to cope with the increasing number of population. The problems of land

management, water, food, and shelter are increasing in the country day by day. Though the population has been more than doubled in the last fifty five years after the liberation, the food production has increased three times as it was before, and the country has no big food crisis. But there is still the possibility of crop failure because of the water pollution and water crisis as well. So, all should extend hands for a permanent solution of food safety through the proper management of water resource of the country. The measures that should be taken to water management are given below-

- 1. The Preservation of Environment:** The rivers, ponds, canals, pools, marshes, fens, forest and land should be preserved.
- 2. Ensuring Proper Use of Water:** In the dry and winter season, the policy and strategy of stopping the misuse of water should be implemented in the country.
- 3. Dredging Rivers Solve the Navigability Problem:** The rivers of the country are filled-up with silt. Almost every year water gets polluted due to flood in different regions of the country. To get rid of this we need preventive steps-flood control and embankment of which we have to be careful. Many rivers have been dried up. The proper dredging will ensure the flow of water and the necessary supply of water for agriculture and industry will be got from here.
- 4. Excavation of Connecting Canal and Reservoir:** If several reservoirs can be excavated in the northern zone of Bangladesh, it will be possible to fulfill the demand of water in the dry season. The demand of water for producing crops will be fulfilled too. Pisces culture and fish-cultivation will be normal.
- 5. River encroachment and pollution prevention:** A major reason for the disruption of water flow in rivers in Bangladesh is that unscrupulous businessmen and politically influential people illegally encroach on river banks or canals, fill them and build structures, obstructing the natural flow of rivers and canals. Gradually, those rivers, canals or water bodies disappear. Local leadership, dignitaries, teachers, students and civil society must play an active role in stopping all these negative activities against rivers and water bodies.
- 6. Involving local communities and students:** Students are the main active force in society. Therefore, it is possible for local communities and students to prevent water pollution, ensure water management and maximum use of water. In this case, students and teachers can play a role in protecting water and the environment by increasing public awareness through public involvement.

5. Saline-free Water: The soil of the southern zone of the country is salty because of sea water. A thin layer over the soil spoils the fertility and causes crop failure. So, Pisces culture, agriculture and afforestation programmes are being hampered. To maintain the supply of freshwater in these areas, the flow of sweet water from the upstream should be ensured. In addition, the Common River Commission should be implemented to gain Bangladesh's share in the management of common rivers according to the international convention of water diplomacy with neighboring countries. Then water and food security can be strengthened.

6. The Protection of River Erosion: In the rainy season, new shore emerges and the river beds are filled-up with silts. The water safety should be ensured by dredging and protecting the erosion of rivers.

7. Using Limited Fertilizers and Pesticides: In our country, fertilizers and pesticides have been indiscriminately used in the agricultural sectors for a long time. So, the cultivation of Pisces culture and agriculture is hindered for acute water pollution. If the indiscriminate use of chemicals can be stopped, the qualities of soil and water can be protected.

8. The Proper Use of Science and Technology: The water resource of the country should be used in the welfare of the human beings. The scientific concepts and the technology should be used in increasing the production of agriculture. If the water resource of the country is ensured, the agricultural production will be increased gradually after fulfilling the need of water of the whole year. That is why we should ensure water and food security, and we should make the 'National Water Policy' properly effective in the country.

The Classification of Forests of Bangladesh

Where the trees grow in plenty is called the forest. Timber, honey, bee-wax, etc. are the resources of the forest. Bangladesh does not have sufficient forests. A country should have 20-25% of forest of the total area of land. But in Bangladesh, it is only 17%. The increasing population of the country requires valuable timber to make house and furniture. The requisite timber is collected from the land of forest of the country. As a result, the forest is gradually being decreased.

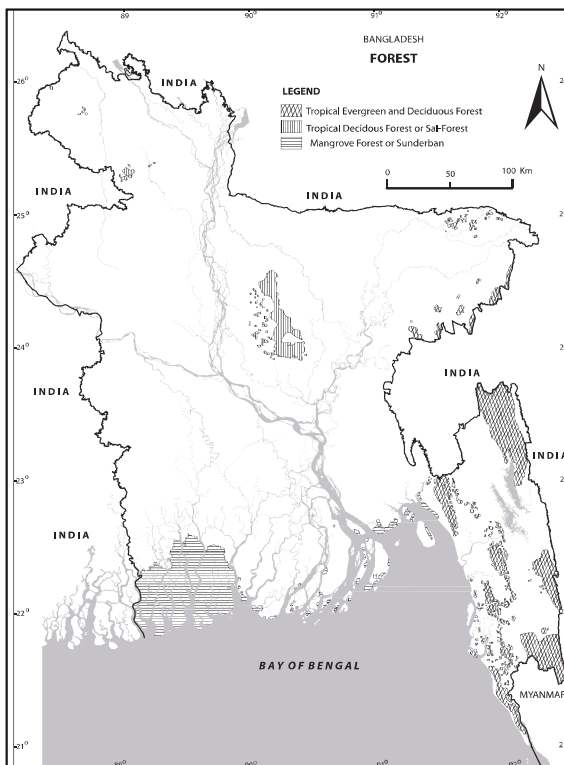
Different types of forests are created in the different areas of Bangladesh on the basis of the climate and the variety of soil. The forest areas of Bangladesh are divided into four distinctive classes the forest zone -of Chattogram, the forest Zone of Sylhet, the Sundarbans, and the forest zone of Dhaka-Tan- gail-Mymensingh. As per the characteristics of trees, forests have been classified as follows: 1. The Tropical Evergreen and Deciduous Forest; 2. The Tropical Deciduous Forest; and 3. The Mangrove or *Goran* Forest.

1. The Tropical Evergreen and Deciduous Forest: The hilly regions of the southeast and north-east of Bangladesh are known as the tropical evergreen and

deciduous forest area. Usually various kinds of trees such as the creeping plants, shrubs, and herbs grow in the hot and moist areas.

The leaves of these trees do not sprout and fall down at a time. As a result, the forest remains green throughout the year. That is why these forests are called evergreen forest. Chattogram, Khagrachori, Rangamati, Bandarban and Sylhet are included in this zone. The total area of this forest is about 14 thousand square kilometers.

The *chapalish*, *moina*, *telsur*, *mehogani*, *jarul*, *segun*, *gorjan* are the mentionable trees of this region. Bamboos and canes grow in plenty in the hilly region of Sylhet. Rubber is cultivated in Chittagong and Sylhet zone too.



Picture 5 3 • The forest land of Bangladesh

2. The Tropical Deciduous Forest: Mymensingh, Tangail, Gazipur, Dinajpur and Rangpur comprise the tropical deciduous forest area. In the forests of these districts, leaves of the trees fall once in a year in the Winter. Besides the *shal* or *gozari*, the *koroi*, *bohera*, *hezol*, *shiris*, *horotoki*, *jackfruit*, *nim* etc. trees grow in plenty. In this forest, as the *shal* is the main tree, it is called the *shal forest*. In Mymensingh, Tangail and Gazipur, this forest is known as the *Modhupur Vowal forest*, and in Dinajpur, it is called the forest land of Borendra.

3. The Mangrove or Goran Forest Area: The trees that grow at the salty and wet soil washed by the ebb and flow of the sea near the seashore of Khulna (southwest zone), Noakhali and Chattogram to the southeast zone of Bangladesh are called the Mangrove or Goran forest area. It grows in the Sundarban too. On the wet and salty soil, the *sundari*, *gewya*, *poshor*, *dhundal*, *kewra*, *bayen*, *goran*, *golpata* etc. trees grow in plenty.

In the national life of Bangladesh, the forest land is important not only for forest resources but also for leading a healthy life in the midst of light, wind and in the green nature.

The Importance of Natural Resources in Economy of Bangladesh

In the economic development of Bangladesh, the importance of natural resources knows no bound. Land, forests, Pisces culture, minerals, solar energy, water etc. are the important natural resources of the country. In the economy of Bangladesh, it is possible to alleviate poverty, ensure food security and standardize life style. These natural resources accelerate the economic development. The land is the most important resource of Bangladesh. The soil of this country is so fertile that the farmers do not need much money for cultivating crops. By ensuring the utmost use of this fertile land, we can extend the production of agricultural crops, flower, and trees. The production of crops has been increased three times during the last forty years after the liberation of Bangladesh. By using the modern technology, high quality seeds, and scientific system of cultivation, Bangladesh may produce more crops. By cultivating various native and foreign fruits, it is possible to fulfill the demand of nutrition, and after fulfilling the local demand of vegetables, it might be exported to foreign country. The use of fertile land of Bangladesh has been increased in a large scale to meet the demands of housing, mills and factories, culverts, roads, towns, cities for the increasing population that has been developed financially as well. Unless the land is used in a planned way, the catastrophe in the national life is inevitable. In our economy, the importance of water as a natural resource is especially mentionable. The economic development of our agriculture and industry completely depend on the water of the rivers, canals, pools, marshes and fens. The communication system depends on the natural resource like water too. The country is going to be solvent economically by using the solar energy, mineral and forest resources of the country. The major portion of the national income of the country comes from the use of these resources. The natural resources of the country contribute a lot to the development of the mills and industries. As a result, the people are getting opportunity for employment, commodities of the local demands are being produced, and the export-oriented goods are being produced by using these resources. In the economic development, new efforts are being adopted. The use of

Group work: Mention the importance of natural resources in the economic development of Bangladesh.

natural resources is helping to take those efforts ahead. The demand and supply of the natural resources are being increased as raw materials, those are being treated with importance, and after fulfilling the demand of the country, these commodities are being exported to the foreign countries. As a result, the economic condition of the people is being developed. People are being more interested and conscious in producing crops, using forests and other natural resources. In this way, the economy of the country is rapidly progressing through the use of the natural resources and the economic efforts of human beings, and the people are moving towards a developed life.

Exercise

Multiple Choice Questions

1. Where is the origin of the Matamuhuri River?
 - a. Mayvar Mountains
 - b. Lushai Hills
 - c. Manasarovar
 - d. Gangotri Glacier
2. The characteristics of Gajhari trees are:
 - i. All leaves fall during different seasons
 - ii. Their leaves remain evergreen
 - iii. They grow in saline soil

Which of the following is correct?

- | | |
|--------------|---------------|
| a. i | b. i and ii |
| c. i and iii | d. ii and iii |

Read the following passage and answer questions 3 and 4.

Sajib went on a study tour to a forest in the eastern region of India and noticed that the trees there were quite tall. His teacher told them that such trees grow in abundance in a region of Bangladesh.

3. Which tree grows in the forest seen by Sajib?

a. Teak	b. Bahera
c. Shirish	d. Dhundul
4. Where in Bangladesh is a similar forest found?

a. Tangail	b. Dinajpur
c. Chittagong Hill Tracts	d. Noakhali

Creative Questions

1. Zahid studies at the University of Chittagong. During his vacation, he takes his foreign classmates to visit the local forest. The beauty of the forest, adorned with Teak, Garjan, and Jarul trees, captivates them. On their way back, Zahid takes them to the bank of the region's main river and tells them that their river is a source of infinite power.

- a. Where is the Naf River located?
- b. Explain the reason for the formation of the Brahmaputra's distributaries.
- c. Explain the characteristics of the forest described in the passage.
- d. Analyze the economic importance of the river mentioned in the prompt.

2. Ajmal Mia was a resident of a river bank in the northern region of the country. There was a time when he used to earn his livelihood by fishing. But currently, due to the changed nature of the river, he has had to change his profession. Having lost his homestead, he has become destitute. In his region, extreme water scarcity in different seasons disrupts public life.

- a. What is a Mangrove or Tidal Forest?
- b. Why is abundant solar energy available in South Asian countries?
- c. Explain the reason for the changed nature of the river in the region where Ajmal Mia resides.
- d. What kind of steps can be taken to resolve the crisis mentioned in the passage? Give your opinion.

Short Answer Questions

1. What is solar resource?
2. What are the advantages of hydroelectric power generation?
3. How would you explain the concept of river conservation?